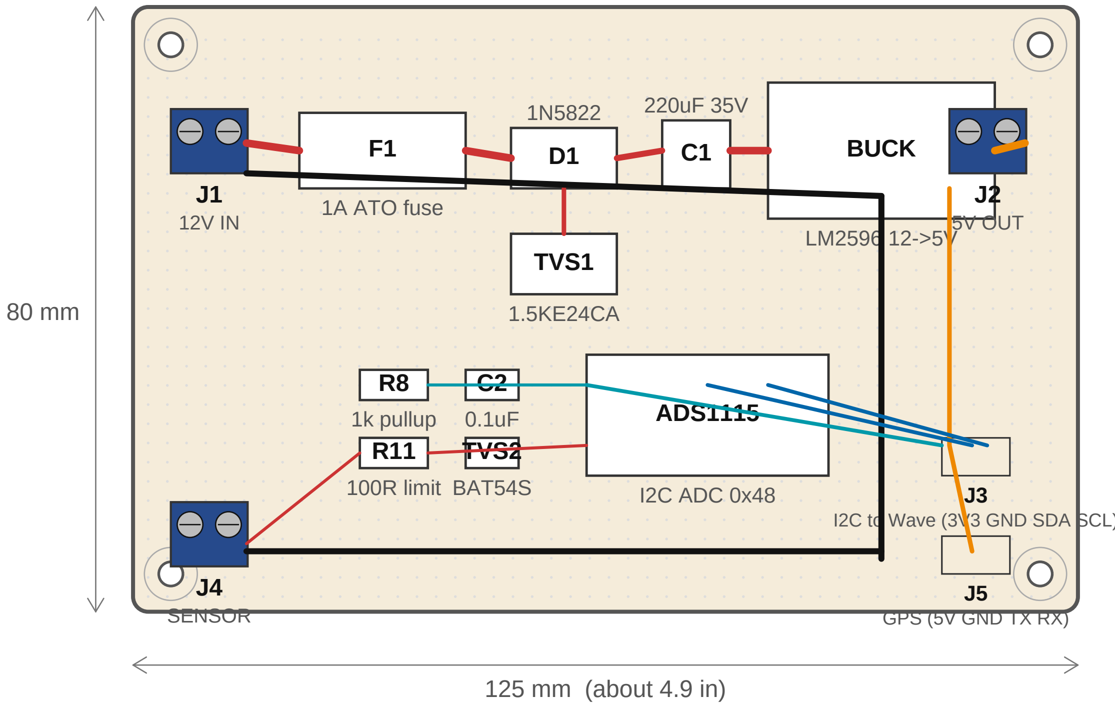
Page 1 of 4 -- Single-Board Hand-Build (Perfboard)

125 x 80 mm prototype perfboard. 2.54 mm hole grid shown. Component placement is the assembly map -- traces shown are wires

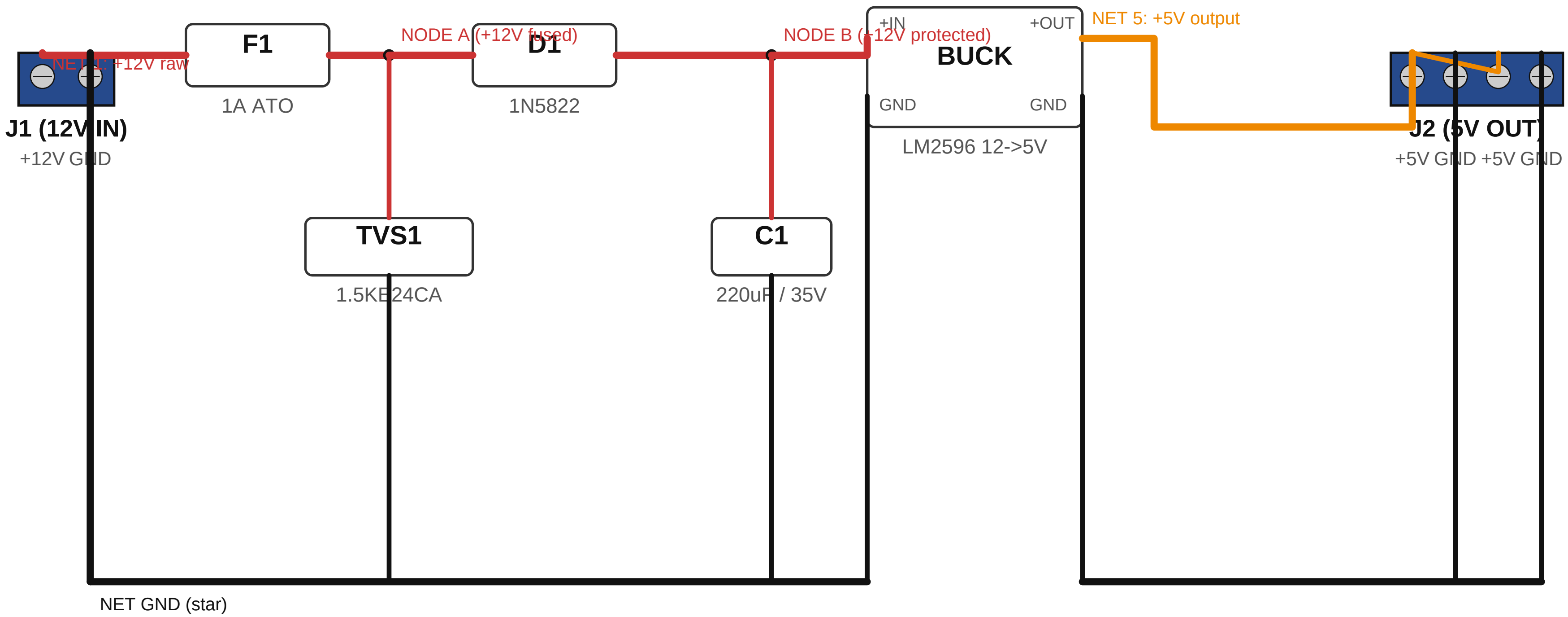


Trace conventions:

+12V	1.0 mm trace -- 1A continuous
+5V	1.0 mm trace -- 1A continuous
+3.3V	0.5 mm trace -- low current to ADC
I2C	0.5 mm trace -- digital signal
Sense	0.4 mm trace -- low-current analog
GND	0.8 mm; star ground at one common point

Page 2 of 4 -- Board A: Power Board (point-to-point)

Schematic-style flow. Each net traceable end-to-end with zero wire crossings. Use as the netlist for EasyEDA.

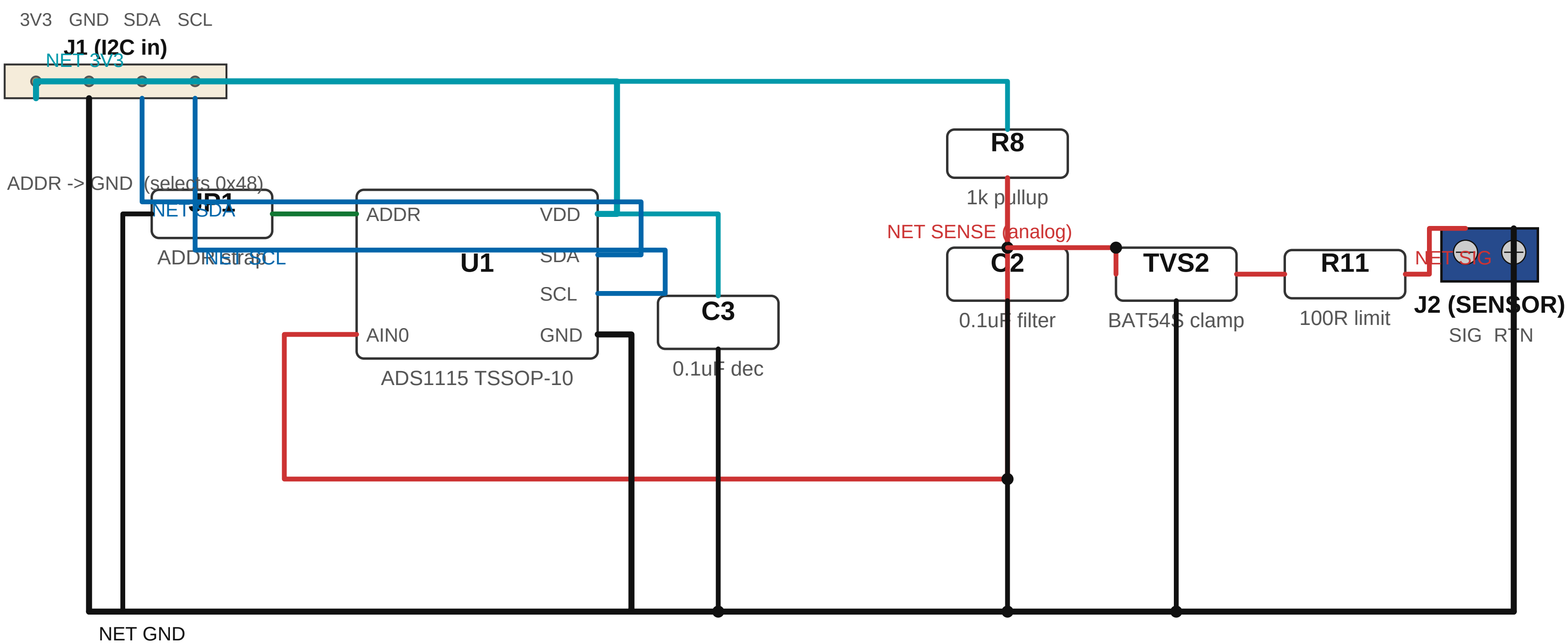


NETLIST -- read each row as one wire / copper trace:

NET 1 (+12V raw)	J1.+12V -> F1.in
NODE A (+12V fused)	F1.out -> TVS1.top -> D1.in
NET 2 (TVS1 shunt)	TVS1.bot -> GND rail
NODE B (+12V protected)	D1.out -> C1.top -> BUCK.+IN
NET 3 (C1 shunt)	C1.bot -> GND rail
NET 5 (+5V output)	BUCK.+OUT -> J2.+5V (pins 1 and 3)
NET GND	J1.GND -> TVS1.bot -> C1.bot -> BUCK.GND -> J2.GND (pins 2 and 4)

Page 3 of 4 -- Board B: Sensor Board (point-to-point, bare-chip)

Schematic-style flow. Bare ADS1115 (TSSOP-10) with C3 decoupling and ADDR-to-GND strap. No wire crossings.



Component sourcing for the hand-build single-board AND the EasyEDA two-board option.

BILL OF MATERIALS (combined; same parts for hand-build or split design)

Ref	Description	Notes
F1	Fuse, 1 A slow-blow ATO + PCB-mount holder	Littelfuse 0287000.PXCN or generic
D1	Diode, 1N5822 Schottky (40 V / 3 A)	axial DO-201
TVS1	TVS, 1.5KE24CA bidirectional	axial; load-dump clamp
C1	Cap, 220 uF / 35 V / 105 deg C electrolytic	radial, 8 mm dia
BUCK	DC-DC buck module, 12V to 5V, 1A min	LM2596 / MP1584 sub-module
ADS1115 (HB)	ADC, ADS1115 breakout module	hand-build single-board only -- header pins
U1 (EDA)	ADC, ADS1115 bare chip TSSOP-10	EasyEDA Sensor Board -- JLCPCB C37593
C2	Cap, 0.1 uF ceramic (X7R, 50 V)	sense-node noise filter
C3 (EDA)	Cap, 0.1 uF ceramic, 0805 X7R	U1 VDD decoupling -- EasyEDA only
TVS2	TVS, BAT54S dual Schottky	SOT-23, low-leakage 3V3 clamp
R8	Resistor, 1 kohm 1% 1/4 W metal-film	pullup on SENSE node
R11	Resistor, 100 ohm 1% 1/4 W metal-film	series current limit
J1 (12V IN)	Screw terminal, 5.08 mm 2-position	Phoenix MPT or generic; rising-cage clamp
J2 (5V OUT)	Screw terminal, 3.5 mm 4-position	two paralleled rails (Wave + GPS)
J3 (I2C)	JST PH 4-pin connector (2.0 mm pitch)	matches Waveshare I2C JST
J4 (SENSOR)	Screw terminal, 5.08 mm 2-position	shielded sender wire
J5 (GPS)	JST 4-pin connector or 4-wire pigtail	+5V / GND / TX / RX
Stand-offs	M3 brass standoffs + screws (4x)	Power+Sensor stack on split design
Perfboard	125 x 80 mm prototype PCB	ELEGOO / Adafruit, 2.54 mm pitch

ORDERING TIPS

* DigiKey or Mouser is cheapest for resistors, caps, diodes, and connectors at any quantity. Amazon is fast for hobby-quantity ADS1115 breakouts and BUCK modules.

* For EasyEDA / JLCPCB fabrication, the 5-board minimum is ~$5 for a 100 mm board. Their PCB Assembly service can populate the small SMD parts (C2, TVS2 SOT-23) cheaply if you use their part library; leave bigger parts (BUCK, ADS1115, screw terminals) as through-hole for hand-soldering after the bare boards arrive.

* Buy ATO fuse holders that mount with two solder pads -- the PCB clip-style ones vibrate loose in cars over time.

* For screw terminals, get 5.08 mm pin pitch with 'rising cage' clamps (wire goes IN the side hole, not under a plate). They hold stranded automotive wire much better and don't cut individual strands when tightened.

* If you ever need to replace the BUCK with a 12V -> 3.3V regulator (skipping the Waveshare's USB path entirely), use a TPS54302 or LM2575 -- pin-compatible with most LM2596 modules but trims to 3.3V.